

Student Exam Submission

Answer to Question 1:

Agile is an iterative methodology where requirements and solutions evolve through collaboration between cross-functional teams. It is highly flexible. Waterfall is a linear sequential design approach where progress flows downwards like a waterfall through phases of Conception, Initiation, Analysis, Design, Construction, Testing, and Deployment.

Advantages of Agile: Flexible to changes, frequent delivery of working software.

Disadvantages of Agile: Hard to predict final costs and timeline.

Advantages of Waterfall: Clear requirements upfront, easy to manage and track.

Disadvantages of Waterfall: Very rigid, hard to adapt if requirements change mid-way.

Example for Agile: A new mobile app startup where user feedback dictates features.

Example for Waterfall: Building software for a medical device or an airplane where safety and strict regulatory compliance require all designs to be finalized upfront.